

LAB # 2

Good practices of programming

OBJECTIVE

Implementing good code practices and code optimization techniques.

Lab Task:

1. Create a design for the mark sheet by taking runtime value of student name, total marks, obtained marks and calculate its percentage, grade and GPA. Use good practices of programming that we have studied and ensure that the outcomes should be presented in a proper Viewable approach.

```
package marksheet;
import java.util.Scanner;

public class MarkSheet {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Total Marks: ");
        double total = sc.nextDouble();

        System.out.print("Enter Obtained Marks: ");
        double obtained = sc.nextDouble();
        double percent = (obtained / total) * 100;
        String grade;
        double gpa;

        if (percent >= 85) { grade = "A+"; gpa = 4.0; }
        else if (percent >= 75) { grade = "A"; gpa = 3.7; }
        else if (percent >= 65) { grade = "B"; gpa = 3.0; }
        else if (percent >= 55) { grade = "C"; gpa = 2.0; }
        else if (percent >= 45) { grade = "D"; gpa = 1.0; }
        else { grade = "F"; gpa = 0.0; }

        System.out.println("\n----- MARKSHEET -----");
        System.out.println("Student Name: " + name);
        System.out.println("Total Marks: " + total);
        System.out.println("Obtained Marks: " + obtained);
        System.out.println("Percentage: " + percent + "%");
        System.out.println("Grade: " + grade);
        System.out.println("GPA: " + gpa);
    }
}
```

Output:

```
run:
Enter Student Name: Affan
Enter Total Marks: 500
Enter Obtained Marks: 420

----- MARKSHEET -----
Student Name: Affan
Total Marks: 500.0
Obtained Marks: 420.0
Percentage: 84.0%
Grade: A
GPA: 3.7
BUILD SUCCESSFUL (total time: 11 seconds)
```

2. Create a class `Rectangle` with attributes `length` and `width`, each of which defaults to 1. Provide methods that calculate the rectangle's perimeter and area. It has set and get methods for both `length` and `width`. The set methods should verify that `length` and `width` are each floating-point numbers larger than 0.0 and less than 20.0. Write a program to test class `Rectangle`.

Code:

```
package rectangle;
import java.util.Scanner;

public class Rectangle {
    double length = 1, width = 1;

    void setValues(double l, double w) {
        if (l > 0 && l < 20) length = l;
        if (w > 0 && w < 20) width = w;
    }

    double area() { return length * width; }
    double perimeter() { return 2 * (length + width); }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Rectangle r = new Rectangle();

        System.out.print("Enter length: ");
        double l = sc.nextDouble();
        System.out.print("Enter width: ");
        double w = sc.nextDouble();
        r.setValues(l, w);

        System.out.println("\nArea = " + r.area());
        System.out.println("Perimeter = " + r.perimeter());
    }
}
```

Output:

```
run:
Enter length: 5
Enter width: 3

Area = 15.0
Perimeter = 16.0
BUILD SUCCESSFUL (total time: 8 seconds)
```